[scale=0.82]
[font=, black!80] at (14.0,20.5) Expansion Rotor Bank Angle & Pitch Depower Risk;
[shift=(0,2.0)] [black!1, rounded corners=6pt] (-0.5,-0.5) rectangle (13.5,17.5); [font=, black!70] at (6
[shift=(5.0,11.5)] [thick] (-3.5,0) – (3.5,0) node[font=, below] ground;
[gray!60, line width=1pt] (0,0) – (4.5,2.6) node[font=, above right] shaft $30°$;
[blue!60, line width=2pt] (3.6,2.1) – (5.0,2.9) node[font=, above] hub rotor;
[cyan!70] (1.8,1.04) circle (2.5pt); [cyan!60!black, line width=1.8pt] (1.8,1.04) – (1.0,1.85); [cyan!60!b
[font=, cyan!70!black] at (0.3,0.6) expansion
rotor;
[-¿, red!60, line width=1pt] (1.6,1.3) ++(0.3,0.15) arc (35:80:0.6); [font=, red!60] at (0.8,1.6) $45°$ bar
[-¿, blue!50, line width=2pt] (-1.5,2.0) – (0.1,0.8); [font=, blue!50] at (-1.8,2.3) wind;
[shift=(1.2,2.0)] [font=, black!60] at (1.8,2.2) Blade section:;
[black!70, line width=1.6pt, rotate around=-45:(0,0)] (0,0) to[out=10, in=180] (2.2,0.35) to[out=5, i
[-¿, blue!60, line width=1.5pt] (-2.2,0.7) – (-0.1,0.0); [font=, blue!60] at (-2.2,1.0) app. wind;
[-¿, red!70!black, line width=1.8pt] (0.9,0.7) – (2.0,2.0); [-¿, red!50, line width=1.2pt, dashed] (0.9,0.
[font=, green!50!black, fill=green!5, draw=green!40!black, rounded corners=3pt, inner sep=3pt] at (
[font=, align=left, fill=black!3, draw=gray!30, rounded corners=4pt, inner sep=5pt] at (10.0,3.5) E
• Hub at ring position on shaft
• Blades banked $45°$ outward
• Sweep hollow truncated cone
• Lift $\rightarrow F_r$ (spreads ring)
$+ F_a$ (shaft compression);
[shift=(16,2.0)] [red!2, rounded corners=6pt] (-0.5,-0.5) rectangle (13.5,17.5); [font=, red!50!black] a
[shift=(5.0,11.5)] [thick] (-3.5,0) – (3.5,0) node[font=, below] ground;
[gray!60, line width=1pt] (0,0) – (2.6,3.1) node[font=, above right] shaft $50°$;
[blue!60, line width=2pt] (1.9,2.3) – (3.2,3.3) node[font=, above] hub rotor;
[red!70] (1.0,1.2) circle (2.5pt); [red!60!black, line width=1.8pt] (1.0,1.2) – (0.3,2.0); [red!60!black, lin
[font=, red!70!black] at (-0.5,0.8) expansion
rotor;
[-¿, red!50, line width=2pt] (-0.8,3.5) – (0.4,1.5); [font=, red!50] at (-1.2,3.8) wind;
[font=, red!70, fill=red!3, draw=red!50, rounded corners=2pt, inner sep=2pt] at (3.5,1.0) BACK-W
[shift=(1.2,2.0)] [font=, red!50!black] at (1.8,2.2) Back-winded:;
[black!70, line width=1.6pt, rotate around=-45:(0,0)] (0,0) to[out=10, in=180] (2.2,0.35) to[out=5, i
[-¿, red!60, line width=1.5pt] (1.3,2.2) – (0.2,0.2); [font=, red!60] at (1.5,2.5) wind from above;
[-¿, red!70!black, line width=1.8pt] (0.9,0.5) – (-0.1,-0.6); [-¿, red!50, line width=1.2pt, dashed] (0.9,
[font=, red!70!black, fill=red!5, draw=red!50!black, rounded corners=3pt, inner sep=3pt] at (1.8,-1.
[font=, align=left, fill=red!3, draw=red!40, rounded corners=4pt, inner sep=5pt] at (10.0,3.5) Back
• Shaft tilts up during depower
• Blade orientation unchanged
• Apparent wind hits from above
• Lift reverses $\rightarrow F_r$ pulls inward
• r_{eff} drops, shaft buckles;
[red!6, draw=red!40!black, line width=1.2pt, rounded corners=4pt] (-0.5,0.0) rectangle (29.5,1.6); [fo